Xianhao Carton Liu

Carton22
liu03008@umn.edu
carton22.github.io

Education

Computer Science Ph.D. Candidate — ***University of Minnesota, Twin Cities***
Advisor: ***Zhu-Tian Chen*** — *Aug 2024 – Now*
Research topic: Human AI Interaction in Physical Environments and Real World

B.Eng. with Honors in Artificial Intelligence (Turing Class) — ***Zhejiang University***
Outstanding honored undergraduate in ***Chu Kochen Honors College*** — *Sep 2020 – June 2024*

Publications

Generative Proxy: Synthesizing Proxy-Based Interfaces for Real-World Interaction Across AR Glasses
Accepted in UIST 2026 [Paper] — *Jun 2026*
- Xianhao Carton Liu, William Chastek, Eric J Gonzalez, Mar Gonzalez-Franco, Chen Zhu-Tian

Can AR Embedded Visualizations Foster Appropriate Reliance on AI in Spatial Decision Making? A Comparative Study of AR X-Ray vs. 2D Minimap
Accepted in CHI 2026 [Paper] [Video] [Github] — *Jan 2026*
- Xianhao Carton Liu, Difan Jia, Tongyu Nie, Evan Suma Rosenberg, Victoria Interrante, Chen Zhu-Tian

Reality Proxy: Fluid Interactions with Real-World Objects in MR via Abstract Representations
Accepted in UIST 2025 [Paper] [Video] — *May 2025*
- Xiaoan Liu, Difan Jia, Xianhao Carton Liu, Mar Gonzalez-Franco, Chen Zhu-Tian

Understanding User Reliance on AI with Eye-Tracking Smart Glasses in Procedural Tasks
Under review in CHI 2027 — *Sep 2026*
- Xianhao Carton Liu, Bingkun Xie, Ruiheng Wang, Zhiyuan (Ryan) Wang, Chen Zhu-Tian

Research Experience

Harvard Visual Computing Group, Harvard University — ***Cambridge, USA***
Research intern advised by ***Zhu-Tian Chen*** *and* ***Hanspeter Pfister*** — *Jan 2024 - May 2024*

Stanford HCI Group, Stanford Univeristy — ***California, USA***
Research intern advised by ***Jackie Yang*** *and* ***James Landay*** — *Jul 2023 - Sep 2023*

ZJU International Design Institute, Zhejiang Univeristy — ***Hangzhou, China***
Research intern advised by ***Lingyun Sun*** *and* ***Zejian Li*** — *Mar 2022 - Jun 2023*

Projects

WebXR Fundamental 3D Interaction Methods for Object Manipulation — ***WebXR for 3D interaction***
[Video] [Github] — *Spring 2024*
- A WebXR interaction framework for object manipulation methods including translation, rotation and resizing.

Generative XR Spatial Interfaces for Human Robot Co-exploration — ***Situated XR Interfaces***
[Video] [Github] — *Summer 2025*
- Reconstructing and synthesizing real-world scene via mobile robot on real-time for human space interaction

Handbot———Apple WWDC2023 Swift Student Challenge — ***Human-centered GenAI***
[Video] [Github] — *Apr 2023*
- A swift playground encouraging critical thinking and editing on LLM and other genAI content in multi-modalities.

Is raining threes worth it?———NBA 3-pointer Data Visualization — ***Data Visualization***
[Video] [Website] — *Dec 2022*
- An interactive visualization website featuring NBA 3-pointers data and key players, received 17000+ views

Skills

Programming	Agentic engineering/WebXR/Unity3D/ROS/Eye Tracking and EEG/AI Model Training
HCI Research	Quantitative Analysis/Interview Thematic Analysis/Participatory Design and Analysis
Devices	Smart glasses and XR headsets/Embedded Computers (Raspi and ESP32)/Turtlebot